

UNIVERSITY PARTNER



UNIVERSITY OF
WOLVERHAMPTON



HERALD
COLLEGE
KATHMANDU

Artificial Intelligence and Machine Learning

(6CS012)

Assignment - III

Group Name : Mask
Group Members : Dibash Thapa Magar (2329199)
Module Leader : Mr. Siman Giri
Tutor : Mr. Rahul Parajuli
Word Count : 832
Submission Date : 12th May, 2025

Table of Contents

1. Question 1	1
Unsupervised Learning Opportunities at eSewa.....	1
Customer Behavior Clustering for Personalization	1
Fraud Pattern Detection.....	1
2. Question 2.....	2
2.1 Overfitting vs. Underfitting in Machine Learning	2
Why It's Problematic:	2
2.2 Neural Networks vs. Autoencoders.....	3
Applications of Autoencoders.....	3

1. Question 1

Unsupervised Learning Opportunities at eSewa

As Nepal's leading digital payment platform, eSewa can leverage unsupervised learning to uncover hidden patterns in user behavior and improve operational efficiency. Two high-impact areas are:

Customer Behavior Clustering for Personalization

Problem

eSewa's user base includes diverse individuals using the platform for various services—mobile top-ups, utility payments, ticket bookings, and merchant transactions. However, treating all users the same limits the effectiveness of marketing and feature offerings.

Approach

Clustering algorithms such as **K-Means** or **Gaussian Mixture Models (GMM)** can segment users based on transaction frequency, payment categories, time of use, and amount spent.

Business

Value

Identified segments (e.g., "utility payers," "frequent shoppers," "light users") enable targeted promotions, personalized offers, and in-app feature prioritization. For example, "frequent utility payers" can be encouraged to set up auto-payments, improving retention and user satisfaction.

Fraud Pattern Detection

Problem

As transaction volumes grow, manually identifying fraudulent activity becomes unsustainable. Traditional rule-based systems may miss novel fraud patterns or produce too many false positives.

Approach

Anomaly detection techniques such as **Isolation Forests** or **Autoencoders** can flag

unusual transactions based on behavioral deviations without requiring labeled fraud data.

Business

Value

Flagged transactions can be queued for review, reducing financial risk. For example, if a typically low-spending user suddenly makes a large transaction at midnight from a new device, the system can hold it for verification. This proactive defense enhances platform trust, reduces fraud losses, and improves regulatory compliance.

Conclusion

By applying unsupervised learning to customer segmentation and fraud detection, eSewa can deliver smarter personalization and robust security—driving growth, trust, and user engagement.

2. Question 2

2.1 Overfitting vs. Underfitting in Machine Learning

Overfitting occurs when a model learns the training data too well, including noise and outliers, resulting in poor generalization to new data. It performs well on training data but poorly on unseen data. For example, a complex polynomial curve that passes through every training point—even noise—may overfit, leading to high variance.

Underfitting happens when a model is too simple to capture the underlying pattern in the data, resulting in poor performance on both training and test sets. For instance, fitting a straight line to a clearly curved relationship would underfit due to high bias.

Why It's Problematic:

- Overfitting leads to models that lack generalization, making them unreliable in real-world scenarios.
- Underfitting results in models that miss important patterns, rendering them ineffective.

Example:

Imagine predicting house prices. An overfit model might memorize every detail of the training houses but fail to predict new prices. An underfit model might only consider

the number of rooms, ignoring factors like location or size, leading to inaccurate estimates.

Both reduce a model's predictive accuracy, making it crucial to find the right balance through techniques like cross-validation, regularization, or model complexity tuning.

2.2 Neural Networks vs. Autoencoders

A normal neural network maps input to a target output using supervised learning (e.g., predicting house prices). It learns to minimize the error between predictions and known labels.

An autoencoder, on the other hand, is an unsupervised neural network that learns to reconstruct its input. It consists of an encoder (compresses input into a latent representation) and a Neural Networks vs. Autoencoders

A normal neural network maps input to a target output using supervised learning (e.g., predicting house prices). It learns to minimize the error between predictions and known labels.

An autoencoder, on the other hand, is an unsupervised neural network that learns to reconstruct its input. It consists of an encoder (compresses input into a latent representation) and a decoder (reconstructs the original input from this compressed form).

Example

In a normal neural network, input might be customer data and the output is churn prediction. In an autoencoder, both input and output are the same customer data, and the model learns the compressed internal structure.

Applications of Autoencoders

One key use case is anomaly detection. For example, in fraud detection for eSewa, an autoencoder is trained on normal transaction patterns. When it encounters a fraudulent transaction, the reconstruction error is high, flagging it as an anomaly.

How They Help

Autoencoders capture the essential structure of normal data. Anything that deviates significantly (like fraud or system faults) results in poor reconstruction, making it an effective unsupervised method for detecting rare or unknown patterns.

(Reconstructs the original input from this compressed form).

Example

In a normal neural network, input might be customer data and the output is churn

prediction. In an autoencoder, both input and output are the same customer data, and the model learns the compressed internal structure.

Applications of Autoencoders:

One key use case is anomaly detection. For example, in fraud detection for eSewa, an autoencoder is trained on normal transaction patterns. When it encounters a fraudulent transaction, the reconstruction error is high, flagging it as an anomaly.

How They Help:

Autoencoders capture the essential structure of normal data. Anything that deviates significantly (like fraud or system faults) results in poor reconstruction, making it an effective unsupervised method for detecting rare or unknown patterns.